

● MEDIUM

● ALGEBRA

● ANSWER KEY

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# Algebra

# 05

10 answers · Rationale-rich · Teach from doc

**Q1****C**

C. The sum of the lengths, in centimeters, of the three sides of the triangle

Choice C is correct. It's given that in triangle  $QRS$ , sides  $QR$  and  $RS$  each have a length of  $x$  centimeters. Therefore, the expression  $2x$  represents the sum of the lengths, in centimeters, of sides  $QR$  and  $RS$ . It's also given that side  $SQ$  has a length of  $y$  centimeters. Therefore, the expression  $2x + y$  represents the sum of the lengths, in centimeters, of sides  $QR$ ,  $RS$ , and  $SQ$ . Since  $2x + y$  is the sum of the lengths, in centimeters, of the three sides of the triangle and  $2x + y = 37$ , it follows that 37 is the sum of the lengths, in centimeters, of the three sides of the triangle.

Choice A is incorrect. The difference, in centimeters, between the lengths of sides  $QR$  and  $SQ$  is  $x - y$ , not 37.

Choice B is incorrect. The difference, in centimeters, between the lengths of sides  $QR$  and  $RS$  is  $x - x$ , or 0, not 37.

Choice D is incorrect. The length, in centimeters, of one of the two sides of equal length is  $x$ , not 37.

**Q2****B**

B.  $4w + 12$

Choice B is correct. It is given that the width of the dance floor is  $w$  feet. The length is 6 feet longer than the width; therefore, the length of the dance floor is  $w + 6$ . So the perimeter is  $w + w + (w + 6) + (w + 6) = 4w + 12$ .

Choice A is incorrect because it is the sum of one length and one width, which is only half the perimeter. Choice C is incorrect and may result from using the formula for the area instead of the formula for the perimeter and making a calculation error. Choice D is incorrect because this is the area, not the perimeter, of the dance floor.

**Q3****39**

The correct answer is 39. It's given that for the linear function  $j$ ,  $m$  is a constant and  $j12 = 18$ . Substituting 12 for  $x$  and 18 for  $jx$  in the given equation yields  $18 = m12 + 144$ . Subtracting 144 from both sides of this equation yields  $-126 = m12$ . Dividing both sides of this equation by 12 yields  $-10.5 = m$ . Substituting  $-10.5$  for  $m$  in the given equation,  $jx = mx + 144$ , yields  $jx = -10.5x + 144$ . Substituting 10 for  $x$  in this equation yields  $j10 = -10.5(10) + 144$ , or  $j10 = 39$ . Therefore, the value of  $j10$  is 39.

**Q4****24**

The correct answer is 24. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is  $y = \frac{1}{3}x - 14$ . Substituting  $\frac{1}{3}x - 14$  for  $y$  in the second equation in the given system yields  $\frac{1}{3}x - 14 = -x + 18$ . Adding 14 to both sides of this equation yields  $\frac{1}{3}x = -x + 32$ . Adding  $x$  to both sides of this equation yields  $\frac{4}{3}x = 32$ . Multiplying both sides of this equation by  $\frac{3}{4}$  yields  $x = 24$ .

**Q5****A**

A.  $25x + 62y \leq 5,440$

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Choice A is correct. It's given that a truck can haul a maximum of 5,630 pounds. It's also given that during one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. It follows that the truck can haul at most  $5,630 - 190$ , or 5,440, pounds of crates. Since  $x$  represents the number of 25-pound crates, the expression  $25x$  represents the weight of the 25-pound crates. Since  $y$  represents the number of 62-pound crates,  $62y$  represents the weight of the 62-pound crates. Therefore,  $25x + 62y$  represents the total weight of the crates the truck can haul. Since the truck can haul at most 5,440 pounds of crates, the total weight of the crates must be less than or equal to 5,440 pounds, or  $25x + 62y \leq 5,440$ .

Choice B is incorrect. This represents the possible combinations of the number of 25-pound crates,  $x$ , and the number of 62-pound crates,  $y$ , the truck can haul during one trip if it can haul a minimum, not a maximum, of 5,630 pounds.

Choice C is incorrect. This represents the possible combinations of the number of 62-pound crates,  $x$ , and the number of 25-pound crates,  $y$ , the truck can haul during one trip if only crates are being hauled.

Choice D is incorrect. This represents the possible combinations of the number of 62-pound crates,  $x$ , and the number of 25-pound crates,  $y$ , the truck can haul during one trip if it can haul a minimum, not a maximum, weight of 5,630 pounds and only crates are being hauled.

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**Q6****C**

C.  $y = -x - 8$

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Choice C is correct. An equation of a line can be written in the form  $y = mx + b$ , where  $m$  is the slope of the line and  $b$  is the  $y$ -intercept of the line. The line shown passes through the point  $(0, -8)$ , so  $b = -8$ . The line shown also passes through the point  $(-8, 0)$ . The slope,  $m$ , of a line passing through two points  $(x_1, y_1)$  and  $(x_2, y_2)$  can be calculated using the equation  $m = \frac{y_2 - y_1}{x_2 - x_1}$ . For the points  $(0, -8)$  and  $(-8, 0)$ , this gives  $m = \frac{-8 - 0}{0 - (-8)}$ , or  $m = -1$ . Substituting  $-8$  for  $b$  and  $-1$  for  $m$  in  $y = mx + b$  yields  $y = -1x - 8$ , or  $y = -x - 8$ . Therefore, an equation of the graph shown is  $y = -x - 8$ .

Choice A is incorrect. This is an equation of a line with a slope of  $-2$ , not  $-1$ .

Choice B is incorrect. This is an equation of a line with a slope of  $1$ , not  $-1$ .

Choice D is incorrect. This is an equation of a line with a slope of  $2$ , not  $-1$ .

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**Q7****C****C. 8**

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields  $8x + 24 = ax + 3$ . Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as  $8x + 3 = ax + 3$ . The two sides of this equation are equivalent when  $a = 8$ . Therefore, if the given equation has infinitely many solutions, the value of  $a$  is 8.

Alternate approach: If the given equation,  $4x + 12 = \frac{ax+3}{2}$ , has infinitely many solutions, then both sides of this equation are equal for any value of  $x$ . If  $x = 0$ , then substituting 0 for  $x$  in the given equation yields  $4(0) + 12 = \frac{a(0)+3}{2}$ , or  $12 = \frac{3}{2}a$ . Dividing both sides of this equation by  $\frac{3}{2}$  yields  $8 = a$ .

Choice A is incorrect. If the value of  $a$  is 0, the given equation is equivalent to  $4x + 12 = 0$ , which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of  $a$  is 3, the given equation is equivalent to  $4x + 12 = \frac{3x+3}{2}$ , or  $4x + 12 = \frac{3}{2}x + \frac{3}{2}$ , which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of  $a$  is 12, the given equation is equivalent to  $4x + 12 = \frac{12x+3}{2}$ , or  $4x + 12 = 6x + 1.5$ , which has one solution, not infinitely many solutions.

**Q8****B****B. 2**

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Choice B is correct. The value of  $f(9) - f(1)$  can be calculated by finding the values of  $f(9)$  and  $f(1)$ . The value of  $f(9)$  can be found by substituting 9 for x in the given function:

$f(9) = \frac{(9+7)}{4}$ . This equation can be rewritten as  $f(9) = \frac{16}{4}$ , or 4. Then, the value of  $f(1)$

can be found by substituting 1 for x in the given function:  $f(1) = \frac{(1+7)}{4}$ . This equation can be

rewritten as  $f(1) = \frac{8}{4}$ , or 2. Therefore,  $f(9) - f(1) = 4 - 2$ , which is equivalent to 2.

Choices A, C, and D are incorrect and may result from incorrectly substituting values of x in the given function or making computational errors.

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**Q9****D****D. 86**

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Choice D is correct. It's given that a wire with a length of 106 inches is cut into two parts. It's also given that one part has a length of  $x$  inches and the other part has a length of  $y$  inches. This can be represented by the equation  $x + y = 106$ . It's also given that the value of  $x$  is 6 more than 4 times the value of  $y$ . This can be represented by the equation  $x = 4y + 6$ . Substituting  $4y + 6$  for  $x$  in the equation  $x + y = 106$  yields  $4y + 6 + y = 106$ , or  $5y + 6 = 106$ . Subtracting 6 from each side of this equation yields  $5y = 100$ . Dividing each side of this equation by 5 yields  $y = 20$ . Substituting 20 for  $y$  in the equation  $x = 4y + 6$  yields  $x = 4(20) + 6$ , or  $x = 86$ .

Choice A is incorrect. This value represents less than half of the total length of 106 inches; however,  $x$  represents the length of the longer part of the wire, since it's given that the value of  $x$  is 6 more than 4 times the value of  $y$ .

Choice B is incorrect. This value represents less than half of the total length of 106 inches; however,  $x$  represents the length of the longer part of the wire, since it's given that the value of  $x$  is 6 more than 4 times the value of  $y$ .

Choice C is incorrect. This represents a part that is 6 more than the length of the other part, rather than 6 more than 4 times the length of the other part.

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# Q10

A

A.

| $x$ | $y$ |
|-----|-----|
| 2   | 19  |
| 4   | 30  |
| 6   | 41  |

Choice A is correct. In each choice, the values of  $x$  are 2, 4, and 6. Substituting the first value of  $x$ , 2, for  $x$  in the given inequality yields  $y > 42 + 8$ , or  $y > 16$ . Therefore, when  $x = 2$ , the corresponding value of  $y$  must be greater than 16. Of the given choices, only choice A is a table where the value of  $y$  corresponding to  $x = 2$  is greater than 16. To confirm that the other values of  $x$  in this table and their corresponding values of  $y$  are also solutions to the given inequality, the values of  $x$  and  $y$  in the table can be substituted for  $x$  and  $y$  in the given inequality. Substituting 4 for  $x$  and 30 for  $y$  in the given inequality yields  $30 > 44 + 8$ , or  $30 > 24$ , which is true. Substituting 6 for  $x$  and 41 for  $y$  in the given inequality yields  $41 > 46 + 8$ , or  $41 > 32$ , which is true. It follows that for choice A, all the values of  $x$  and their corresponding values of  $y$  are solutions to the given inequality.

Choice B is incorrect. Substituting 2 for  $x$  and 8 for  $y$  in the given inequality yields  $8 > 42 + 8$ , or  $8 > 16$ , which is false.

Choice C is incorrect. Substituting 2 for  $x$  and 13 for  $y$  in the given inequality yields  $13 > 42 + 8$ , or  $13 > 16$ , which is false.

Choice D is incorrect. Substituting 2 for  $x$  and 13 for  $y$  in the given inequality yields  $13 > 42 + 8$ , or  $13 > 16$ , which is false.

## Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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